
The Hidden Link between RLHF and Contrastive Learning

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Abstract

Alignment of large language models (LLMs) with human values has garnered significant attention, with prominent examples including Reinforcement Learning from Human Feedback (RLHF) and Direct Preference Optimization (DPO). In this work, we demonstrate that these alignment objectives can be usefully interpreted through the lens of mutual information (MI) maximization, establishing an approximate but practical connection to contrastive learning. Specifically, we show that methods like DPO leverage the Donsker–Varadhan (DV) lower bound on MI. This analytical perspective identifies a concrete gradient-starvation failure mode in DV-style objectives: they struggle to provide effective updates when the base model’s probability of generating the correct answer is close to zero, thereby limiting the elicitation of complex reasoning capacities. To address this limitation, we introduce Mutual Information Optimization (MIO) as a targeted estimator substitution. By replacing the DV bound with the Jensen–Shannon (JS) MI estimator, MIO structurally overcomes this optimization bottleneck. Comprehensive theoretical analysis and extensive empirical evaluations demonstrate that MIO mitigates the late-stage decline in chosen-likelihood observed in DPO, achieving competitive or superior performance across various benchmarks. The code is available at our [GitHub repository](#).

1. Introduction

Recently, large language models (LLMs) have attained state-of-the-art performance on a wide array of natural-language tasks (Bai et al., 2022). Aligning their outputs with human value is commonly achieved through the methodology

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of reinforcement learning from human feedback (RLHF) (Ouyang et al., 2022)(Stiennon et al., 2022), whose fine-tuning recipe has become one of the industry standards. However, its dependence on two step reinforcement learning presents challenges (Xiao et al., 2025), such as computational inefficiency and training instability. To mitigate these limitations, alternative one-step approaches such as direct preference optimization (DPO) (Rafailov et al., 2023) and its variants (Meng et al., 2024)(Tajwar et al., 2024) have been proposed, which has successfully eliminated the need for explicit reward modeling. Specifically, an implicit reward based on the likelihood of preference data is defined, which results in significant gains in efficiency while preserving competitive performance.

Major Questions

1. *Can RLHF-style preference optimization be formalized through contrastive MI surrogates?*
2. *When do DV/MINE-based objectives provide weak learning signals for low-probability responses?*
3. *Can a targeted MI-estimator substitution improve the stability and performance of preference optimization?*

Several recent works have hinted at conceptual links between RLHF and imitation learning (Ho & Ermon, 2016). DIL (Xiao et al., 2025) shows that optimizing a policy with RLHF implicitly increases the likelihood of human-preferred outputs, and formalizes one connection between language-model alignment and imitation learning. IRL (Wulfmeier et al., 2024) examines pretraining and fine-tuning from an imitation-learning standpoint, arguing that standard next-token fine-tuning can be viewed as imitation learning. Beyond imitation learning, prior work has also drawn parallels between DPO and contrastive learning (Jaiswal et al., 2021; Xu et al., 2024a; Chen et al., 2024). We do not claim that this broad connection is entirely new. Instead, our goal is to make a specific DV/MINE-based formalization precise enough to connect reward learning, DPO, and KL-regularized RLHF, and then use that formalization to motivate a practical estimator substitution.

Recent empirical studies have challenged the widely held belief that *reinforcement learning with variable rewards* (RLVR) or standard KL-regularized RLHF can continuously unlock genuinely novel reasoning trajectories. Yue et al. (2025) observe that, on benchmarks such as GSM8K and

MATH, reasoning chains produced after RLHF often remain within the sampling support of the *base* model, suggesting that RL may primarily *amplify* pre-existing capabilities rather than discover new ones. Our main focus is RLHF-style subjective preference optimization rather than verifier-based RLVR, but this observation motivates a closely related optimization question. Prior studies also report that DPO can collapse when $\pi_\theta(y^- | x)$ becomes very small. We therefore ask: **why can DV/MINE-based preference objectives provide weak updates for rare but valuable responses, and why does DPO suffer chosen/rejected likelihood collapse when the rejected-response probability becomes extremely small?**

RLHF and its variants fundamentally adhere to a reward maximization objective, often instantiated through parametric models such as the Bradley–Terry (BT) model (Bradley & Terry, 1952). This modeling paradigm, while effective in aligning with preference data, has been shown to suffer from overfitting (Yuan et al., 2024; Pal et al., 2024), frequently leading to suboptimal generalization and misalignment with true user preferences (Xu et al., 2024b; Wu et al., 2024). Recent empirical studies further indicate that DPO and its variants progressively focus on unlearning the rejected responses. This process inadvertently increases the model’s tendency to generate out-of-distribution responses, rather than reinforcing alignment with the chosen responses (Xu et al., 2024b). From a theoretical perspective, Yan et al. (2025) analyzes the optimization dynamics of DPO and demonstrates that the shared tokens between chosen and rejected responses lead to gradient interference. This interference introduces instability and a coupled degradation, whereby suppressing rejected responses inadvertently reduces the model’s confidence in the selected responses. Such simultaneous unlearning of both rejected and chosen outputs ultimately degrades downstream performance—particularly in reasoning-intensive and mathematical tasks (Pal et al., 2024; Yuan et al., 2024; Meng et al., 2024). This observation motivates our third research question: **Can we design a principled algorithm that overcomes inherent shortcomings of DPO, while effectively enhancing model performance on complex reasoning and mathematical tasks?**

This paper makes the following contributions. (1) We present an approximate mutual-information framework for RLHF-style preference optimization, showing how reward-model training, DPO, and KL-regularized RLHF can be interpreted through DV/MINE-based contrastive surrogates under explicit assumptions. (2) We characterize the main approximation gaps and make their boundedness visible in the main text rather than leaving the connection implicit. (3) We identify a DV-induced gradient-starvation mechanism that explains why low-probability responses receive weak effective updates and why DPO can exhibit

late-stage synchronous degradation of chosen and rejected likelihoods. (4) We propose MIO, a JSD-based replacement for the DV/MINE surrogate, and evaluate it on math, reasoning, broader alignment, online preference-optimization, and adapted GRPO/GSPO comparisons while also discussing sensitivity and label-noise tradeoffs.

2. Rethinking RLHF: A Contrastive Learning Perspective

2.1. RLHF-style Preference Optimization via Contrastive MI Surrogates

Answer to Question 1: RLHF can be viewed through contrastive mutual-information (MI) surrogates.

Setup and notation. Let $x \sim \mathcal{D}$ be a user prompt. For each x , we observe two candidate responses: a *chosen* response y^+ preferred by humans and a *rejected* response y^- with lower preference. We aim to fine-tune a language model (policy) $\pi_\theta(y|x)$ so that it assigns higher likelihood to y^+ than to y^- , while remaining close to a reference model $\pi_{\text{ref}}(y|x)$.

For analysis, we assume $y^+ \sim \pi_{\text{chosen}}(\cdot|x)$ and $y^- \sim \pi_{\text{rejection}}(\cdot|x)$, where both π_{chosen} and $\pi_{\text{rejection}}$ are energy-based re-weightings of π_{ref} (see Appendix A). We also define the response mixture pool $\bar{\pi}(\cdot|x) = \frac{1}{2}\pi_{\text{chosen}}(\cdot|x) + \frac{1}{2}\pi_{\text{rejection}}(\cdot|x)$.

We further reformulate this problem as a **contrastive learning objective**:

$$\max_{\theta} I(\pi_\theta, \pi_{\text{chosen}}) \text{ and } \min_{\theta} I(\pi_\theta, \pi_{\text{rejection}}) \quad (1)$$

We begin by considering the unconstrained maximization of mutual information with respect to π_{chosen} , which reduces to a standard **imitation learning problem**:

$$\max_{\theta} I(\pi_\theta, \pi_{\text{chosen}}) \quad (2)$$

In Appendix B, we demonstrate that the contrastive learning and imitation learning objectives are equivalent, as improving one objective decreases the other and the strict local maximum of the former corresponding to the strict local minimum of the latter.

Since the mutual information cannot be directly computed, we introduce the Donsker–Varadhan (DV) / MINE lower bound (Belghazi et al., 2018a). Let $T_\phi : \mathcal{X} \times \mathcal{Y} \rightarrow \mathbb{R}$ be a scalar-valued critic (implemented as a neural network). For fixed ϕ , $T_\phi(x, y)$ is deterministic.

The standard DV/MINE form is

$$I = \sup_{\phi} \mathbb{E}_{P_{XY}}[T_\phi(x, y)] - \log \mathbb{E}_{P_X P_Y} \left[e^{T_\phi(x, y)} \right]. \quad (3)$$

In our setting, the *positive* joint distribution is induced by

$$(x, y^+) \sim P_{XY}^+ \quad \text{where} \quad x \sim \mathcal{D}, \quad y^+ \sim \pi_{\text{chosen}}(\cdot|x).$$

Hence, we can derive the following expression for equation 2 (see Appendix C.1):

$$I = \sup_{\phi} \left[\mathbb{E}_{x \sim \mathcal{D}, y^+ \sim \pi_{\text{chosen}}(\cdot|x)} [T_{\phi}(x, y^+)] - \log \left(\mathbb{E}_{x \sim \mathcal{D}} \left[\mathbb{E}_{y \sim P_Y} \left[e^{T_{\phi}(x, y)} \right] \right] \right) \right]. \quad (4)$$

Where P_Y denotes the response marginal (exactly computing it is intractable). To obtain a practical objective with explicit negatives, we lower-bound the marginal term by using the mixture $\bar{\pi}$:

$$I \geq \sup_{\phi} \left[\mathbb{E}_{x \sim \mathcal{D}, y^+ \sim \pi_{\text{chosen}}(\cdot|x)} [T_{\phi}(x, y^+)] - \log \left(\mathbb{E}_{x \sim \mathcal{D}} \left[2 \mathbb{E}_{y \sim \bar{\pi}(\cdot|x)} \left[e^{T_{\phi}(x, y)} \right] \right] \right) \right]. \quad (5)$$

The right-hand side of equation 5 is commonly referred to as the modified DV bound I_{DV} (Hjelm et al., 2019).

Consider a mini-batch with M chosen pairs $\{(x_i, y_i^+)\}_{i=1}^M$ and N rejected pairs $\{(x_j, y_j^-)\}_{j=1}^N$. We estimate the expectation by Monte Carlo estimator:

$$\begin{aligned} & \mathbb{E}_{x \sim \mathcal{D}, y \sim \bar{\pi}(\cdot|x)} \left[e^{T_{\phi}(x, y)} \right] \\ & \approx \frac{1}{2} \left(\frac{1}{M} \sum_{i=1}^M e^{T_{\phi}(x_i, y_i^+)} + \frac{1}{N} \sum_{j=1}^N e^{T_{\phi}(x_j, y_j^-)} \right). \end{aligned} \quad (6)$$

Substituting equation 6 into equation 5 yields the empirical objective:

$$\begin{aligned} & \max_{\phi} \hat{I}_{\text{DV}}(\phi) = \\ & \log \left(\frac{e^{T_{\phi}(x_i, y_i^+)}}{\frac{1}{M} \sum_{m=1}^M e^{T_{\phi}(x_m, y_m^+)} + \frac{1}{N} \sum_{n=1}^N e^{T_{\phi}(x_n, y_n^-)}} \right) \end{aligned} \quad (7)$$

When $M = N = 1$, equation 7 simplifies to:

$$\max_{\phi} \log \sigma(T_{\phi}(x, y^+) - T_{\phi}(x, y^-)). \quad (8)$$

where T_{ϕ} is equivalent to the reward model in RLHF, see in Appendix A. Then, by treating T_{ϕ} as the reward model, we can derive an objective function that aligns precisely with the reward loss under the BT preference assumption in RLHF.

Furthermore, instead of training a separate neural critic T_{ϕ} , we can restrict the critic to a log-ratio family that depends on θ :

$$T_{\theta}(x, y) = \beta \left(\log \pi_{\theta}(y|x) - \log \pi_{\text{ref}}(y|x) \right), \beta > 0. \quad (9)$$

Substituting equation 9 into equation 8 yields the DPO objective (Rafailov et al., 2023):

$$\max_{\theta} \log \sigma \left(\beta \log \frac{\pi_{\theta}(y^+|x)}{\pi_{\text{ref}}(y^+|x)} - \beta \log \frac{\pi_{\theta}(y^-|x)}{\pi_{\text{ref}}(y^-|x)} \right). \quad (10)$$

Keep constrain the critic to the log-ratio sub-family equation 9. Only substituting this into the right-hand side of the MINE objective equation 4 and applying certain approximations yields the following objective used in the second stage of RLHF, where the policy model is updated using the reward model (Wu et al., 2024; Ghosh et al., 2020):

$$\begin{aligned} & \max_{\theta} \mathbb{E}_{x \sim \mathcal{D}, y \sim \pi_{\theta}(\cdot|x)} [T_{\phi}(x, y)] \\ & - \beta_{\text{KL}} \mathbb{E}_{x \sim \mathcal{D}} \text{KL}(\pi_{\theta}(\cdot|x) \parallel \pi_{\text{ref}}(\cdot|x)), \end{aligned} \quad (11)$$

The approximation error introduced here is upper-bounded (proof in Appendix E), and the bound depends on the ratio $\frac{\pi_{\theta}(\cdot|x)}{\pi_{\text{ref}}(\cdot|x)}$: the smaller this ratio, the smaller the approximation error. This offers us a new and insightful perspective about why PPO introduces a clipping function to constrain the update magnitude of the policy—large deviations in this ratio lead to large approximation errors, which in practice degrade performance. The above derivation successfully transforming the RLHF task into a contrastive learning framework.

Furthermore, we can draw some insights:

1. Reward learning in RLHF can be interpreted as learning a critic function $T \in \mathcal{F}_{\text{all}}$ that estimates the relevant preference-induced MI surrogate (see Appendix C).
2. When the critic is restricted to the log-ratio sub-family $\mathcal{F}_{\text{sub}}$, the policy update admits a contrastive surrogate interpretation: it increases alignment between π_{θ} and the chosen-response distribution while decreasing alignment with the rejected-response distribution. If, instead of training a separate parameterized neural network T_{ϕ} , we restrict the critic search to $\mathcal{F}_{\text{sub}}$, we recover the DPO loss as a particular log-ratio surrogate.

2.2. DV/MINE-style Surrogates Can Induce Gradient Starvation.

Answer to Question 2: When do PPO/DPO-style updates receive weak signals for small $\pi_{\theta}(y^*|x^*)$? A mutual information explanation

In essence, based on our analysis, both RLHF and DPO can be viewed as approximate procedures for optimizing a DV/MINE-based mutual information estimator $I_{\text{DV}}(\theta)$. However, such estimators can provide weak gradient directions for a specific response y^* when the reference policy $\pi_{\text{ref}}(y^* | x^*)$ approaches zero. Since π_{θ} is typically initialized as a frozen copy of the reference model, if the

initial $\pi_\theta(y^* | x^*)$ is close to zero, then no matter how well the reward model is trained, using it to further optimize π_θ becomes ineffective. **One mechanism behind this phenomenon is the DV/MINE surrogate $I_{DV}(\theta)$.** To formalize this claim, we present two theorems:

Theorem 1 (DV/MINE Starvation Theorem). If $\pi_{\text{ref}}(y^* | x^*) = 0$, then the DV/MINE mutual information estimator $I_{DV}(\theta)$ cannot guide any meaningful update for the specific response y^* . Formally,

$$\langle \nabla_\theta I_{DV}(\theta), \nabla_\theta \log \pi_\theta(y^* | x^*) \rangle = 0.$$

This explains why PPO-style methods (e.g., GRPO) typically require a *cold start* stage prior to increase the probability of the base model sampling the correct answer y^* : when the probability of sampling the correct answer y^* is too small, the cold start could increase it to prevent policy gradients from vanishing due to $\pi_{\text{ref}}(y^* | x^*) = 0$.

Some readers may object that, in practice, neural networks assign nonzero probability to every sequence (albeit extremely small), so Theorem 1 may not hold exactly. To address this, we provide a strengthened result:

Theorem 2 (DV/MINE Starvation Theorem Pro). As $\pi_\theta(y^* | x^*)$ approaches zero, the DV/MINE estimator $I_{DV}(\theta)$ yields increasingly misleading gradients. Formally,

$$|\langle \nabla_\theta I_{DV}(\theta), \nabla_\theta \log \pi_\theta(y^* | x^*) \rangle| \leq 2L \pi_\theta(y^* | x^*),$$

where L is a constant. A full proof is provided in Appendix H.

This theorem offers a mutual-information perspective on why off-policy settings in PPO/GRPO often lead to degraded training performance: under an off-policy regime, the probability $\pi_\theta(y | x)$ of sampling a trajectory is extremely small, which makes the gradient of I_{DV} nearly orthogonal to the policy gradient. As a result, the mutual information estimator cannot be used to approximate the gradient of the policy model π_θ . This also provides an intuitive insight: the effectiveness of DV/MINE-based gradient estimation is highly sensitive to the initial values of $\pi_\theta(y | x)$. Once these probabilities are too small, the estimator I_{DV} becomes *too hungry to update*.

In summary, the DV/MINE estimator introduces serious drawbacks: not only does it induce large gradient variance (see in Appendix D), but it also leads to *policy-gradient starvation* whenever $\pi_{\text{ref}}(y | x)$ is near zero. This severely hinders the learning dynamics of RLHF. Mitigation strategies such as cold-start initialization are often used to increase $\pi_{\text{ref}}(y | x)$, thereby reducing gradient starvation. However, the fundamental culprit remains the DV/MINE estimator itself. This motivates an important research question: **are there effective alternatives to DV/MINE-based mutual information estimators?**

3. Methods: Mutual-Information Optimisation

3.1. Let us replace DV/MINE Estimator with JSD in alignment.

As our primary objective is to maximize Mutual Information rather than precisely estimate its value (Hjelm et al., 2019), we can rely on non-KL divergences, which offer favorable trade-offs. For instance, the Jensen-Shannon MI estimator (Nowozin et al., 2016)—similar to the binary cross-entropy used in minimizing total correlation (Brakel & Bengio, 2017)—is well-understood in neural network optimization and is more stable in practice compared to the DV-based objective (Hjelm et al., 2019):

$$I_{\text{JSD}}(\theta) = \sup_{\phi} \left(\mathbb{E}_{x \sim \mathcal{D}, y^+ \sim \pi_{\text{chosen}}(\cdot | x)} [-\text{sp}(-T_\phi(x, y^+))] - \mathbb{E}_{x \sim \mathcal{D}, y^- \sim \bar{\pi}(\cdot | x)} [\text{sp}(T_\phi(x, y^-))] \right). \quad (12)$$

where $\text{sp}(u) = \log(1 + e^u)$. Considering the following Monte Carlo estimator:

$$\mathbb{E}_{x \sim \mathcal{D}, y \sim \bar{\pi}(\cdot | x)} [\text{sp}(T_\phi(x, y))] \approx \frac{1}{2} \left[\frac{1}{M} \sum_{i=1}^M \text{sp}(T_\phi(x_i, y_i^+)) + \frac{1}{N} \sum_{j=1}^N \text{sp}(T_\phi(x_j, y_j^-)) \right] \quad (13)$$

By replacing the DV/MINE (Belghazi et al., 2018a) surrogate I_{DV} in equation 4 with the JS-based surrogate I_{JSD} given in equation 12, and substituting equation 13 into equation 12, we obtain the following empirical objective:

$$\hat{I}_{\text{JSD}}(\phi) = -\frac{1}{M} \sum_{i=1}^M \text{sp}(-T_\phi(x_i, y_i^+)) - \frac{1}{2} \left(\frac{1}{M} \sum_{i=1}^M \text{sp}(T_\phi(x_i, y_i^+)) + \frac{1}{N} \sum_{j=1}^N \text{sp}(T_\phi(x_j, y_j^-)) \right). \quad (14)$$

Setting $M = N = 1$ and constraining the critic to a log ratio family (See in Appendix C.2) gives a *closed-form* loss, which we call **Mutual-Information Optimisation (MIO)**:

$$\mathcal{L}_{\text{MIO}}(\theta) = \log \left[1 + \left(\frac{\pi_{\text{ref}}(y^+ | x)}{\pi_\theta(y^+ | x)} \right)^\beta \right] + \frac{1}{2} \log \left[1 + \left(\frac{\pi_\theta(y^+ | x)}{\pi_{\text{ref}}(y^+ | x)} \right)^\beta \right] + \frac{1}{2} \log \left[1 + \left(\frac{\pi_\theta(y^- | x)}{\pi_{\text{ref}}(y^- | x)} \right)^\beta \right]. \quad (15)$$

3.2. Failure Mode of DPO when $\pi_\theta(y | x)$ is Close to Zero

Essentially, DPO estimates the policy gradient through the DV/MINE mutual information estimator. Consequently, it

suffers from the same failure mode: when $\pi_\theta(\cdot | x)$ becomes too small, the policy gradient vanishes. This effect is particularly pronounced for the rejected sample y^- in preference alignment, since training explicitly encourages reducing its probability, making $\pi_\theta(y^- | x)$ more likely to approach zero over the course of optimization.

For a preference triple (x, y^+, y^-) the DPO loss can be written as (Yan et al., 2025):

$$\ell_{\text{DPO}} = \log(1 + [\alpha z^\beta]), \alpha = \left[\frac{\pi_{\text{ref}}(y^+ | x)}{\pi_{\text{ref}}(y^- | x)} \right]^\beta, z = \frac{\pi_\theta(y^- | x)}{\pi_\theta(y^+ | x)}.$$

Denoting $\pi^+ = \pi_\theta(y^+ | x)$ and $\pi^- = \pi_\theta(y^- | x)$, differentiation gives

$$\frac{\partial \ell_{\text{DPO}}}{\partial \pi^+} = -\frac{\alpha \beta}{1 + \alpha z^\beta} \frac{z^\beta}{\pi^+}, \quad \frac{\partial \ell_{\text{DPO}}}{\partial \pi^-} = \frac{\alpha \beta}{1 + \alpha z^\beta} \frac{z^{\beta-1}}{\pi^-}.$$

Hence, we have that $|\partial_{\pi^-} \ell| / |\partial_{\pi^+} \ell| = \pi^+ / \pi^-$. The gradient on the *rejected* probability therefore grows much faster than that on the *chosen* probability; as $\pi^- \rightarrow 0$ we have $\partial_{\pi^-} \ell \rightarrow \infty$ but $\partial_{\pi^+} \ell \rightarrow 0$. Because chosen and rejected completions usually share many tokens, the vanishing positive gradient cannot offset the dominant negative gradient, so the model is driven to lower *both* probabilities. This synchronous likelihood collapse—observed empirically in late-stage DPO training—degrades overall performance (Pal et al., 2024).

3.3. Stability of the MIO Gradient

Answer to Question 3: MIO leads to a more stable training process!

If the DV/MINE mutual information estimator causes gradient starvation when $\pi_\theta(y | x)$ is small, replacing it with a JSD-based mutual information estimator eliminates this issue. In other words, when $\pi_\theta(y | x)$ approaches zero, MIO does not suffer from vanishing policy gradients. This claim can be verified directly by differentiating the MIO loss. The MIO objective for the same triple is

$$\ell_{\text{MIO}} = \ln(1 + e^{-\beta \text{LR}^+}) + \frac{1}{2} \ln(1 + e^{\beta \text{LR}^+}) + \frac{1}{2} \ln(1 + e^{\beta \text{LR}^-}),$$

with $\text{LR}^\pm = \log \pi_\theta^\pm - \log \pi_{\text{ref}}^\pm$. Writing $\sigma^\pm = \sigma(\beta \text{LR}^\pm)$, one obtains

$$\partial_{\pi^+} \ell_{\text{MIO}} = \frac{\beta}{\pi^+} (1.5\sigma^+ - 1), \quad \partial_{\pi^-} \ell_{\text{MIO}} = \frac{\beta}{2\pi^-} \sigma^-.$$

Proposition 3.1 (Selective suppression of negatives). *Let $\pi^- \rightarrow 0$. Then, we have:*

$$\left| \frac{\partial \ell_{\text{MIO}}}{\partial \pi^-} \right| = \frac{\beta}{2\pi^-} \sigma^- \rightarrow \infty, \quad \left| \frac{\partial \ell_{\text{MIO}}}{\partial \pi^+} \right| = \frac{\beta}{\pi^+} |1.5\sigma^+ - 1|,$$

so the negative likelihood continues to be forcefully suppressed, while the positive gradient remains bounded and therefore *does not vanish*. Consequently, MIO preserves a strong corrective signal for rejected responses without impairing the optimisation of chosen responses.

Proposition 3.2 (Self-regulating positive gradient). *Define $\sigma^+ = \sigma(\beta \text{LR}^+)$. Then, we have:*

$$\partial_{\pi^+} \ell_{\text{MIO}} = \frac{\beta}{\pi^+} (1.5\sigma^+ - 1) \begin{cases} > 0 & \text{if } \sigma^+ > \frac{2}{3}, \\ < 0 & \text{if } \sigma^+ < \frac{2}{3}. \end{cases}$$

Hence, MIO exhibits an *intrinsic clamping mechanism*: once a chosen token attains excessive probability ($\sigma^+ > 2/3$) the gradient turns positive, gently pushing its likelihood downward and mitigating over-fitting; conversely, if the token remains under-represented ($\sigma^+ < 2/3$) the gradient is negative, encouraging further amplification. This self-adjusting, bi-phase behaviour amounts to an implicit *self-paced curriculum*: optimisation effort is automatically re-allocated from already-well-learned positives to the more informative, harder ones, yielding superior robustness and sample efficiency compared with DPO.

4. Experiment

4.1. DPO Can Degrade When $\pi_\theta(y^- | x)$ is Near 0; MIO Mitigates This Effect

The previous section theoretically demonstrated that methods dependent on $I_{\text{DV}}(\theta)$, such as DPO and PPO, encounter issues with policy gradients becoming ineffective when $\pi_\theta(y | x)$ approaches zero, especially for $\pi_\theta(y^- | x)$. This is particularly evident because $\pi_\theta(y^- | x)$ is more likely to approach zero during training than $\pi_\theta(y^+ | x)$. However, this problem does not occur in the MIO approach. The earlier discussion was purely theoretical, and while it is challenging to validate this in practice, we have prepared a toy model experiment to empirically investigate this issue.

Following (Yan et al., 2025), the toy model setup, shown in Figure 1, consists of a discrete space with 4 prompts and 10 responses. The policy π_θ , implemented as a three-layer MLP, processes a one-hot input vector and outputs a categorical distribution over the responses, which are divided into three categories: selected responses (first 4 dimensions), rejected responses (next 4 dimensions), and unseen responses (final 2 dimensions). Each prompt has a corresponding optimal response (e.g., response 1 is optimal for prompt 1). In DPO or MIO training, preference data is generated through a mini-batch sampling strategy, where an ideal annotator matches each prompt with its optimal response, creating a diagonal preference matrix. In each mini-batch, one additional response is randomly selected to form preference pairs, ensuring diverse gradient updates.

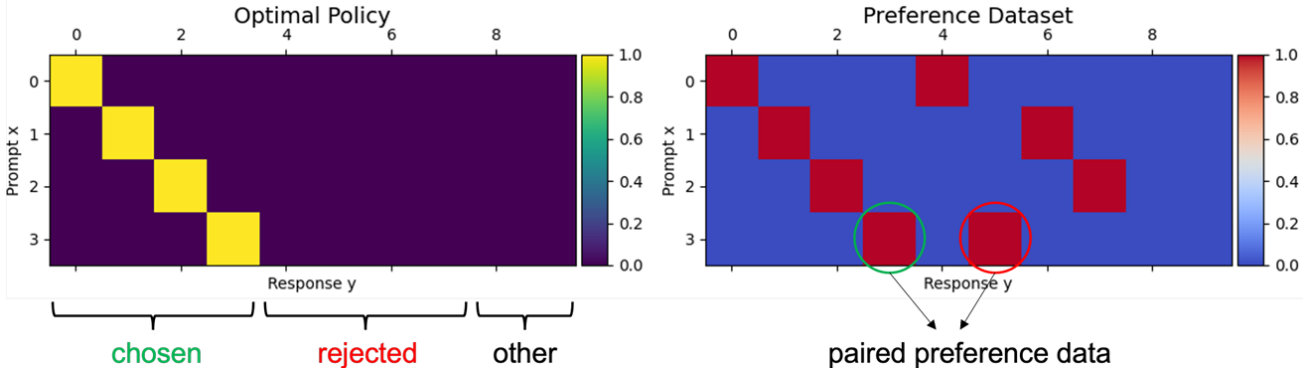


Figure 1. Toy model setup(Yan et al., 2025). Left: the optimal policy where the highlighted blocks represent optimal responses. Right: preference dataset construction.

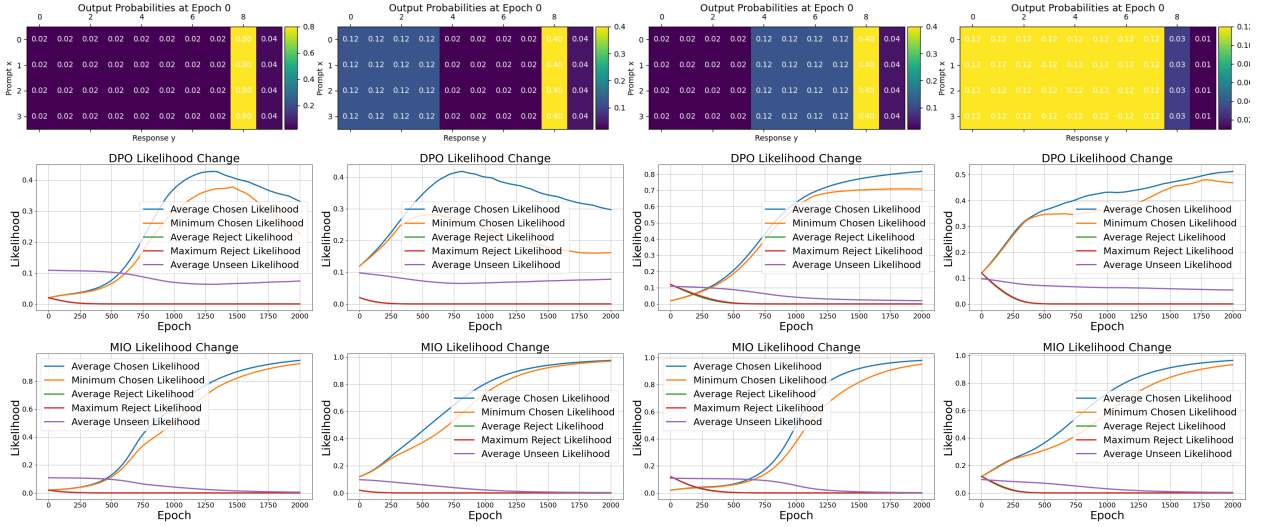


Figure 2. Overview of the DPO and MIO dynamics. From left to right, the figures show the initial state and the likelihood dynamics for chosen/rejected/unseen responses in Scenarios 1 to 4, similar to the left diagram in Figure 2: (1) both chosen and rejected responses are very small, (2) chosen is normal and rejected is very small, (3) chosen is small and rejected is normal, and (4) both chosen and rejected is normal.

Figure 2 contrasts the learning dynamics of DPO with those of our proposed MIO objective. When the probabilities of both the *chosen* and *rejected* responses remain in a moderate range, DPO converges smoothly and the likelihood of the chosen response declines only marginally relative to that of the rejected one. Once the rejected-response probability becomes very small, however, DPO develops a pathological *synchronous collapse*: the probabilities of *both* the chosen and the rejected responses decrease in lock-step. Across all regimes we tested, MIO displays no such joint degradation. When the rejected probability is pushed towards zero, DPO and its derivatives exhibit synchronous decline of the chosen and rejected likelihoods, mirroring what is observed in practice (Yan et al., 2025; Chen et al., 2024; Ho & Ermon, 2016), **whereas MIO does not exhibit this failure mode in our tested settings**. A formal explanation of this qualitative difference is provided in Section 3.2.

The toy model serves as an abstract simulation that amplifies the 3D-properties of DPO (Yan et al., 2025), which leads to a decrease in the likelihood of both chosen and rejected responses. However, MIO does not exhibit this effect. While the toy model setup differs significantly from real-world LLM training—such as in sampling frequency—it provides valuable insights. In real-world scenarios, DPO and MIO are typically trained over one epoch, with each data point used only a few times. In contrast, the toy model samples the same data points repeatedly. Conceptually, this is akin to treating each input/output as a token rather than a full prompt/response, where each token may be sampled multiple times during real-world training.

Since findings from toy experiments may not reliably generalize to the complex dynamics of large-scale language model training, we further conduct a local-update diagnostic on an actual Qwen2.5-7B-Instruct model. Specifically, we partition the training examples into discrete buckets based

Table 1. Local update diagnostic on the Qwen2.5-7B-Instruct model. Examples are bucketed according to the log-probability of the rejected response. Notably, in buckets 0–6 (where the rejected response is highly unlikely), DPO frequently decreases the chosen likelihood, whereas MIO consistently increases it.

Bucket	Rej.	DPO		MIO	
	Log-Prob	Coeff.	Δ Chosen	Coeff.	Δ Chosen
0	−942.91	0.0017	−0.1825	0.0076	3.3761
1	−688.73	0.0026	−0.1905	0.0076	3.1313
2	−602.39	0.0031	−0.2306	0.0074	2.9332
3	−536.32	0.0032	−0.1853	0.0073	2.3901
4	−489.57	0.0037	−0.2391	0.0072	2.8037
5	−447.91	0.0041	−0.0671	0.0070	2.3695
6	−410.26	0.0043	−0.0167	0.0070	2.5174
7	−370.93	0.0044	0.0608	0.0069	2.1924
8	−325.69	0.0047	0.1349	0.0067	1.5627
9	−250.09	0.0049	0.0683	0.0063	1.1715

on the log-probability of their rejected responses. We then compare the effective gradient updates induced by DPO and MIO on the likelihood of the chosen response.

The empirical results presented in Table 1 strongly corroborate our theoretical mechanism. In buckets 0–6, which correspond to instances where the rejected response has an extremely low probability, the DPO update frequently results in a *decrease* in the chosen likelihood, whereas MIO *consistently increases* it. This empirical evidence confirms that the failure mode of DPO is not merely an artifact of sampling frequency. Even when both chosen and rejected responses are present in the diagnostic batch, the inherent gradient coupling in DPO actively suppresses the chosen-response update whenever the rejected response is highly improbable.

4.2. Real World Experimental

Evaluation on reasoning benchmarks. We fine-tune both Mistral-7B-SFT (Tunstall et al., 2023) and LLaMA3-8B-SFT on the 64k-prompt *UltraFeedback-Binarized* (Cui et al., 2024) corpus, as well as train directly on Qwen2.5-7B-Instruct, where each prompt is paired with a *chosen* and a *rejected* completion selected by GPT-4, yielding 64k positive–negative pairs; we retain the official train/validation split of *UltraFeedback-Binarized*.

Model quality is assessed on eight public mathematical and reasoning suites: Hendrycks MATH (Hendrycks et al., 2021a) (12.5 k competition-level problems plus the level-5 “hard” subset), Minerva Math (Lewkowycz et al., 2022a) (quantitative-reasoning items distilled from GSM8K and MATH), MultiMedQA (Singhal et al., 2022) (six medical QA benchmarks spanning USMLE, PubMedQA, and consumer health queries), MathQA (Amini et al., 2019a) (37 k multiple-choice word problems with executable rationales),

GSM8K (Cobbe et al., 2021a) (8.5 k grade-school arithmetic questions), AQuA-RAT (Ling et al., 2017) (100 k algebraic problems with free-text solutions), and MuSR (Sprague et al., 2024) (narrative tasks requiring multistep soft reasoning over 700–1000-word stories). The code, hyperparameter settings, and evaluation metrics is provided in the Appendix G.

The result is in table 2. For Mistral-7B-SFT, MIO achieves the best performance on five tasks and ranks second on two others across the eight mathematical and reasoning benchmarks. For LLaMA3-8B-SFT, MIO ranks first in seven of the eight benchmarks and second in one. For Qwen2.5-7B-Instruct, MIO achieves the best performance across all eight benchmarks. These results demonstrate that MIO offers superior generalization in reasoning and mathematical problem-solving compared to existing alignment methods.

Evaluation on broader alignment benchmarks. We conducted comprehensive evaluations of the **Mistral-7B-SFT** model trained with MIO across core reasoning, math, and coding capabilities, as summarized in Table 3. Notably, MIO achieves substantial gains on Arena-Hard, yielding an absolute improvement of +7.3 (a +44.7% relative increase) over DPO.

In addition to the standard math, coding, and reasoning suites, we further evaluate MIO on a broader spectrum of alignment benchmarks. Specifically, we fine-tune **Mistral-7B-SFT** on the **UltraFeedback-Binarized** dataset and compare MIO against DPO on **MMLU-Pro**, **TruthfulQA** (MC1), **AlpacaEval 2** (win rate), and **MT-Bench** (GPT-4 judge). As shown in Table 3, MIO yields consistent improvements across MMLU, TruthfulQA, and AlpacaEval 2, while successfully matching the performance of DPO on MT-Bench.

Comparison with PPO. To compare MIO with classical RLHF, we follow the **GPT-J SFT** setup used in the DPO paper and train on the **Reddit TL;DR summarization** experiment. We then evaluate **out-of-domain** on the **CNN/DailyMail** test split using win rate against the ground-truth summary at two decoding temperatures.

Table 4. GPT-4 win rates vs. ground truth summaries for out-of-distribution CNN/DailyMail input articles.

	Temp 0 $\uparrow$ Temp 0.25 $\uparrow$	
PPO	0.26	0.23
DPO	0.36	0.31
MIO	0.38	0.33

Table 2. Evaluation results on 8 benchmarks. The **best** and second best are marked.

Model (↓) / Benchmark (→)		Hendrycks Math	Minerva Math	Multimedqa	MathQA	GSM8K	Aqua Rat	Math Hard	MUSR
Mistral-7B-SFT	SFT	0.14	8.80	49.03	29.91	32.29	19.29	2.72	39.94
	DIL(Xiao et al., 2025)	0.04	6.46	49.54	28.94	28.35	21.33	2.11	38.36
	DPO(Rafailov et al., 2023)	0.04	8.14	49.21	30.72	36.47	19.69	2.72	43.65
	IPO(Azar et al., 2023)	0.08	6.50	48.39	29.41	3.71	19.69	0.23	35.71
	KTO(Ethayarajh et al., 2024)	0.04	10.38	<u>49.76</u>	30.56	41.75	21.65	3.43	43.39
	NCA(Chen et al., 2024)	0.14	8.92	49.58	30.59	37.68	18.50	2.49	39.68
	ORPO(Hong et al., 2024)	0.04	<u>9.68</u>	<u>49.40</u>	29.78	36.39	<u>24.80</u>	2.57	42.06
	SimPO(Meng et al., 2024)	<u>0.19</u>	8.36	<u>49.16</u>	<u>30.93</u>	39.20	18.11	2.49	39.02
	SLiC(Zhao et al., 2023)	0.08	8.72	49.74	30.66	39.73	20.47	<u>3.17</u>	42.99
	CPOXu et al. (2024a)	0.16	7.14	49.18	27.57	31.59	23.37	3.40	42.44
MIO	0.21	9.08	49.82	30.96	<u>40.16</u>	24.83	3.57	<u>43.42</u>	
LLaMA3-8B-SFT	SFT	0.06	14.06	56.84	28.07	50.26	25.90	4.61	39.94
	DIL(Xiao et al., 2025)	<u>0.19</u>	17.16	57.01	34.85	<u>55.86</u>	23.62	<u>5.06</u>	40.75
	DPO(Rafailov et al., 2023)	0.04	16.48	55.54	28.17	54.38	23.62	4.23	39.11
	IPO(Azar et al., 2023)	0.12	15.26	56.99	29.85	53.07	25.69	4.91	36.77
	KTO(Ethayarajh et al., 2024)	0.04	16.42	31.36	19.83	55.57	22.05	5.03	35.05
	NCA(Chen et al., 2024)	0.08	<u>17.24</u>	57.53	28.41	54.81	25.20	4.46	38.62
	ORPO(Hong et al., 2024)	0.08	16.86	53.81	28.01	48.98	25.59	4.68	38.76
	SimPO(Meng et al., 2024)	0.06	17.16	57.13	<u>36.35</u>	53.37	24.80	4.61	<u>41.08</u>
	SLiC(Zhao et al., 2023)	0.17	15.18	51.94	28.58	46.22	<u>26.77</u>	3.59	38.03
	CPOXu et al. (2024a)	0.13	14.26	54.01	26.13	52.16	25.39	3.85	39.15
MIO	0.20	17.68	<u>57.32</u>	36.85	56.86	27.62	5.44	42.81	
Qwen-7B-Instruct	Instruct	0.04	25.20	55.54	33.33	71.19	21.65	49.47	38.5
	DIL(Xiao et al., 2025)	0.12	23.12	55.31	33.66	72.21	24.80	46.22	37.29
	DPO(Rafailov et al., 2023)	0.11	26.16	<u>57.36</u>	33.21	73.46	22.83	50.08	39.55
	IPO(Azar et al., 2023)	0.18	25.44	56.28	34.20	71.57	21.26	<u>50.38</u>	<u>40.08</u>
	KTO(Ethayarajh et al., 2024)	<u>0.19</u>	26.91	52.86	29.7	<u>75.08</u>	20.08	46.00	37.99
	NCA(Chen et al., 2024)	0.07	25.34	56.61	34.30	72.56	22.05	49.47	39.81
	ORPO(Hong et al., 2024)	0.02	26.34	55.49	<u>37.98</u>	74.30	<u>28.58</u>	32.18	39.97
	SimPO(Meng et al., 2024)	<u>0.19</u>	26.6	45.25	34.80	71.72	25.98	38.44	37.96
	SLiC(Zhao et al., 2023)	0.06	<u>27.54</u>	57.32	36.31	73.99	25.98	46.07	38.76
	CPOXu et al. (2024a)	0.02	18.76	54.76	36.14	72.71	28.74	46.27	37.17
MIO	0.22	27.84	57.61	38.17	75.39	30.16	51.36	40.34	

Table 3. Comprehensive evaluation results. We report performance across core reasoning, math, and coding capabilities alongside broader alignment benchmarks. The best results for each metric are highlighted in bold.

Method	Math, Reasoning, and Coding							Broader Alignment			
	Arena-Hard	IFEval	BBH	GPQA	HumanEval	TheoremQA	LeetCode	MMLU	TruthfulQA	AlpacaEval 2	MT-Bench
DPO	16.3	43.29	43.27	28.35	31.7	9.8	2.2	28.72	36.35	16.3	7.3
MIO	23.6	45.66	45.63	30.29	35.5	12.6	4.3	28.98	37.97	21.2	7.3


 Figure 3. Training curves on MISTRAL-7B-SFT(per token level). Unlike DPO, IPO, and ORPO, MIO *increases* the rewards of chosen responses while suppressing rejected ones, thereby avoiding the “synchronous collapse” observed in prior methods.

4.3. MIO prevents the chosen-reward collapse

During DPO training, the rewards of *all* responses (y^+ and y^-) often decrease simultaneously, a harmful “unlearning”

effect that degrades downstream performance (Xiao et al., 2025; Meng et al., 2024; Pal et al., 2024). Figure 3 shows that the proposed MIO objective substantially mitigates this failure mode: the chosen-rewards increase while the rejected rewards drop slightly. Section 3.2 provides a theoretical explanation of this behavior.

Additionally, our goal is not to minimize the rejected reward (since y^+ and y^- share many tokens), but rather to maximize the chosen–rejected reward accuracy while avoiding collateral degradation of the chosen output. Overly suppressing the rejected response often, through shared tokens and gradient interference, also pulls down the likelihood/reward of the chosen response and may even trigger out-of-distribution (OOD) behaviors. Therefore, a healthy learning signal is characterized by a moderate decrease in

Table 5. Online preference optimization on TL;DR summarization using Qwen3.5-9B.

Method	Win Rate	Quality
Online DPO	30.1	3.92
Online IPO	28.9	3.86
Online MIO	35.5	4.01

the rejected response together with an increase in the chosen response.

Some readers may argue that NCA-Pair can also increase the chosen reward while decreasing the rejected reward. To examine this, Figure 5 (in Appendix) presents a direct comparison between MIO and NCA-Pair in terms of reward margin and related metrics. The results show that MIO achieves a greater increase in both the chosen reward and the reward margin, as well as a larger decrease in the rejected reward, compared to NCA-Pair. This indicates that MIO leverages the reward signal more effectively than NCA-Pair.

4.4. Online Preference Optimization

Although our theoretical pairwise analysis applies most directly to offline DPO-style optimization, MIO can also be used in online preference optimization by generating candidates from the current policy and assigning preference labels with a reward model or judge.

We evaluate online preference optimization on TL;DR summarization using Qwen3.5-9B. Online MIO outperforms online DPO and online IPO in both win rate and judged quality.

This result suggests that the JSD-style modification is not limited to static offline preference datasets. However, we emphasize that this experiment does not claim to model the full clipped-PPO training dynamics. It only shows that the MIO loss can be used effectively in an online preference-optimization loop.

4.5. Sensitivity and Tradeoffs

We study sensitivity to the coefficient β . Because MIO uses average token log-probability normalization, its β values are not directly comparable to DPO coefficients. Empirically, MIO remains stable over a broad range from 2.0 to 12.5. Smaller values slightly favor instruction-following, while moderately larger values slightly favor reasoning.

MIO is not universally better in every regime. While MIO benefits from a stronger learning signal, this characteristic inherently renders it more sensitive to label noise. To quantify this sensitivity, we trained the Qwen-2.5-7B-Instruct model on the UltraFeedback dataset, artificially introducing noise by randomly flipping 10% of the preference labels in

Table 6. Sensitivity to the hyperparameter β . The best results for each metric are highlighted in bold.

Metric	β						
	2.0	5.0	7.5	10.0	12.5	15.0	20.0
IFEval	46.37	45.96	45.88	45.66	43.38	39.19	36.59
GSM8K	39.91	39.35	39.33	40.16	39.77	39.57	38.74

table 7. On the MUSR benchmark, the baseline Qwen model achieves a score of 38.5. When exposed to this corrupted data, the performance of DPO drops to 36.7, whereas MIO experiences a slightly more pronounced degradation, falling to 36.0. This indicates that because MIO extracts richer signals from the provided labels, it is also more susceptible to fitting erroneous data compared to DPO. Consequently, these findings confirm that the overall efficacy of MIO is contingent upon the availability of high-quality preference data.

Table 7. Performance on the MUSR benchmark under 10% label noise.

Method	MUSR (10% Noise)
Qwen (Baseline)	38.5
DPO	36.7
MIO	36.0

5. Conclusion

In this work, we present an approximate MI-based perspective connecting RLHF-style preference optimization and contrastive learning over the distributions of chosen and rejected responses. Building on this connection, we provide theoretical explanations for several phenomena observed in prior work. For example, reasoning paths that are not sampled by the base model remain inaccessible even after RL training, and DPO can receive weak updates when the probability of a rejected response becomes too small. These issues are consistent with the DV/MINE-induced gradient-starvation mechanism analyzed here. Furthermore, we introduce the **Mutual Information Optimization (MIO)**, replacing the Donsker-Varadhan (DV) bound with a Jensen-Shannon (JSD) bound. MIO often improves preservation of reasoning and mathematical abilities and offers greater training stability compared to existing baselines. Extensive experimental results demonstrate that MIO achieves competitive or improved performance on a range of mathematical and reasoning benchmarks. We hope that our work may offer useful insights for the community and help address the limitations of current RLHF methods. This work does not explore the effects of alternative mutual information estimators, which we leave for future research.

Impact Statement

This paper presents a unifying mutual-information (MI) perspective on aligning large language models (LLMs) with human preferences, interpreting RLHF and DPO as contrastive learning objectives based on positive and negative samples from a base model. Building on this view, we propose Mutual Information Optimization (MIO) using a Jensen–Shannon MI estimator, with the goal of improving stability and effectiveness of preference optimization—particularly for reasoning and mathematical tasks—relative to existing approaches.

If adopted, the methods in this work could yield positive impacts by enabling more reliable and sample-efficient alignment of LLMs, potentially reducing the cost and engineering complexity associated with RLHF-style pipelines. More stable preference optimization may also improve model helpfulness on complex reasoning problems, which can benefit downstream applications in education, scientific assistance, software development, and other domains where correctness and robustness matter.

However, improved alignment and reasoning capability can also increase the utility of LLMs for harmful purposes, such as generating persuasive misinformation, automating abusive content, or assisting in the development of unsafe instructions. In addition, preference optimization methods inherit biases present in preference data, annotator populations, and the choice of base model; optimizing for particular feedback signals may unintentionally reinforce majority viewpoints, over-penalize minority dialects or cultural norms, or yield models that are overly confident, leading to harm in high-stakes settings.

To mitigate these risks, we recommend pairing preference-optimization methods with (i) careful dataset curation and documentation of annotator guidelines and demographics when possible, (ii) evaluation across diverse groups, languages, and domains, including stress tests for hallucinations and overconfidence on reasoning tasks, (iii) red-teaming and misuse-oriented evaluations prior to deployment, and (iv) deployment safeguards such as content filtering, rate limits, and monitoring, especially for applications with security or safety implications. Overall, we believe MIO can provide meaningful benefits when used with appropriate governance, transparency about limitations, and responsible deployment practices. Furthermore, we have released all code associated with training and experiments, which is publicly available at our [GitHub repository](#) for full reproducibility.

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A. Reward Modeling as Critic Learning under the BT Assumption.

Energy-based form of the *chosen* model. Throughout the analysis we assume that the preference-filtered policy π_{chosen} is obtained from a frozen *reference* model π_{ref} via an energy re-weighting:

$$\pi_{\text{chosen}}(y | x) = \pi_{\text{ref}}(y | x) \frac{e^{\tau r(x, y)}}{Z(x)}, \quad (\text{A.1})$$

Definition A.1. For a positive pair (x, y^+) , define the log-ratio critic

$$T_{\theta}^{\text{ratio}}(x, y^+) := \log \frac{\pi_{\theta}(y^+ | x)}{\pi_{\text{chosen}}(y^+ | x)}. \quad (\text{A.2})$$

Assume the reward satisfies

$$r(x, y^+) = \kappa \left(\log \frac{\pi_{\theta}(y^+ | x)}{\pi_{\text{ref}}(y^+ | x)} + \log Z(x) \right), \quad \kappa > 0. \quad (\text{A.3})$$

Substituting (A.3) into (A.1) yields

$$\pi_{\text{chosen}}(y^+ | x) = \pi_{\text{ref}}(y^+ | x)^{1-\tau\kappa} \pi_{\theta}(y^+ | x)^{\tau\kappa} Z(x)^{\tau\kappa-1}. \quad (\text{A.4})$$

Plugging (A.4) into (A.2) gives

$$T_{\theta}(x, y^+) = (1 - \tau\kappa) \left(\log \pi_{\theta}(y^+ | x) - \log \pi_{\text{ref}}(y^+ | x) + \log Z(x) \right). \quad (\text{A.5})$$

In particular,

$$T_{\theta}^{\text{ratio}}(x, y^+) = \log \frac{\pi_{\theta}(y^+ | x)}{\pi_{\text{chosen}}(y^+ | x)} \propto \log \frac{\pi_{\theta}(y^+ | x)}{\pi_{\text{ref}}(y^+ | x)}, \quad (\text{A.6})$$

where the proportionality hides an x -dependent additive constant that cancels in pairwise differences. Define a compact scaling

$$\gamma := \frac{1 - \tau\kappa}{\kappa}. \quad (\text{A.7})$$

Then (A.5) and (A.3) imply

$$T_{\theta}^{\text{ratio}}(x, y^+) = \gamma r(x, y^+), \quad (\text{A.8})$$

and analogously for a negative pair (x, y^-) ,

$$T_{\theta}^{\text{ratio}}(x, y^-) = \gamma r(x, y^-). \quad (\text{A.9})$$

B. Maximizing mutual information towards the positive model inherently distances it from the negative model.

Definition B.1. For every prompt x , draw $y^+ \sim \pi_{\text{chosen}}(\cdot | x)$ and $y^- \sim \pi_{\text{rejection}}(\cdot | x)$. Fix a critic $T_{\phi} \in \mathcal{F}$. With $z^+ = T_{\phi}(x, y^+)$ and $z^- = T_{\phi}(x, y^-)$, define the two DV/InfoNCE estimators:

$$\hat{I}^+ = \log \frac{e^{z^+}}{e^{z^+} + e^{z^-}} = -\log \left(1 + e^{-(z^+ - z^-)} \right), \quad (\text{B.1})$$

$$\hat{I}^- = \log \frac{e^{z^-}}{e^{z^+} + e^{z^-}} = -\log \left(1 + e^{(z^+ - z^-)} \right). \quad (\text{B.2})$$

Write the **logit gap** $\Delta := z^+ - z^-$. With $\sigma(u) = \frac{1}{1+e^{-u}}$ we have $\hat{I}^+ = \log \sigma(\Delta)$, $\hat{I}^- = \log \sigma(-\Delta)$.

Gradients share the Δ direction but opposite sign.

$$\frac{\partial \hat{I}^+}{\partial \Delta} = \sigma(-\Delta), \quad \frac{\partial \hat{I}^-}{\partial \Delta} = -\sigma(\Delta). \quad (\text{B.1})$$

Because $\sigma(u) > 0 \forall u \in \mathbb{R}$,

$$\text{sign}(\partial_{\Delta} \hat{I}^+) = -\text{sign}(\partial_{\Delta} \hat{I}^-). \quad (\text{B.4})$$

Coupled parameter gradients. Let the model parameters be θ and assume z^+, z^- are differentiable in θ . By the chain rule

$$\nabla_{\theta} \hat{I}^+ = \frac{\partial \hat{I}^+}{\partial \Delta} \nabla_{\theta} \Delta, \quad \nabla_{\theta} \hat{I}^- = \frac{\partial \hat{I}^-}{\partial \Delta} \nabla_{\theta} \Delta. \quad (\text{B.5})$$

Combine equation B.1–equation B.5:

$$\nabla_{\theta} \hat{I}^- = -\frac{\sigma(\Delta)}{\sigma(-\Delta)} \nabla_{\theta} \hat{I}^+. \quad (\text{B.6})$$

The scalar factor $-\sigma(\Delta)/\sigma(-\Delta) < 0$; hence

$$\boxed{\langle \nabla_{\theta} \hat{I}^+, \nabla_{\theta} \hat{I}^- \rangle < 0} \quad (\text{B.7})$$

i.e. the two gradient vectors always point in **opposite directions**.

Consequence (first-order). Any update $\theta \leftarrow \theta + \eta \nabla_{\theta} \hat{I}^+$ ($\eta > 0$) that **increases** the positive MI estimator must **decrease** the negative MI estimator:

$$\hat{I}^+(\theta_{\text{new}}) > \hat{I}^+(\theta) \implies \hat{I}^-(\theta_{\text{new}}) < \hat{I}^-(\theta). \quad (\text{B.8})$$

Second-order coupling: extrema coincide

Shared critical points. From equation B.6, $\nabla_{\theta} \hat{I}^+ = 0$ iff $\nabla_{\theta} \hat{I}^- = 0$; thus both objectives share the same set of stationary points.

Hessian analysis. Let $f(\theta) = \hat{I}^+(\theta) = h(\Delta(\theta))$ and $g(\theta) = \hat{I}^-(\theta) = h(-\Delta(\theta))$ with $h(u) = \log \sigma(u)$. Then

$$h'(u) = \sigma(-u) > 0, \quad h''(u) = -\sigma(u)\sigma(-u) < 0.$$

Using standard composition rules,

$$\nabla_{\theta}^2 f = h''(\Delta) \nabla \Delta \nabla \Delta^{\top} + h'(\Delta) \nabla^2 \Delta, \quad (\text{B.9})$$

$$\nabla_{\theta}^2 g = h''(-\Delta) \nabla \Delta \nabla \Delta^{\top} - h'(-\Delta) \nabla^2 \Delta. \quad (\text{B.10})$$

At any common critical point θ^* we have $\nabla \Delta(\theta^*) = 0$; the rank-one terms vanish, leaving

$$\nabla_{\theta}^2 f(\theta^*) = h'(\Delta^*) \nabla^2 \Delta(\theta^*), \quad \nabla_{\theta}^2 g(\theta^*) = -h'(-\Delta^*) \nabla^2 \Delta(\theta^*). \quad (\text{B.11})$$

Because $h'(\cdot) > 0$, the two Hessians differ *only* by an overall sign.

Extremum equivalence. If $\nabla_{\theta}^2 f(\theta^*) \prec 0$ (negative definite), then $\nabla_{\theta}^2 g(\theta^*) \succ 0$ (positive definite); hence

$$\boxed{\begin{aligned} &\theta^* \text{ is a } \textit{strict} \text{ local maximum of } \hat{I}^+ \\ \iff &\theta^* \text{ is a } \textit{strict} \text{ local minimum of } \hat{I}^-. \end{aligned}} \quad (\text{B.12})$$

Implication for optimisation. Therefore,

$$\arg \max_{\theta} \hat{I}^+(\theta) = \arg \min_{\theta} \hat{I}^-(\theta),$$

provided the optimum is strict and isolated. Gradient-based methods that converge to a strict maximiser of $\hat{I}^+$ will, by construction, converge to the corresponding strict minimiser of $\hat{I}^-$, rigorously validating the coupling intuition expressed in the main text.

C. Mutual-Information Estimation: Practical Surrogate and Constraining the Critic T

C.1. DV / MINE Sampling Scheme for RLHF

Sampling notation. Let $x \sim \mathcal{D}$ be a prompt. A chosen completion is denoted $y^+ \sim \pi_{\text{chosen}}(\cdot|x)$ and a rejected completion is denoted $y^- \sim \pi_{\text{rejection}}(\cdot|x)$. We define the mixture pool

$$\bar{\pi}(\cdot|x) = \frac{1}{2}\pi_{\text{chosen}}(\cdot|x) + \frac{1}{2}\pi_{\text{rejection}}(\cdot|x).$$

A critic (reward model) is a scalar function $T_\phi : \mathcal{X} \times \mathcal{Y} \rightarrow \mathbb{R}$ that scores (x, y) .

Modified DV bound. With the above sampling, the DV/MINE bound and its mixture pool relaxation can be written as

$$I(\pi_\theta, \pi_{\text{chosen}}) = \sup_{\phi} \mathbb{E}_{x \sim \mathcal{D}, y^+ \sim \pi_{\text{chosen}}(\cdot|x)} [T_\phi(x, y^+)] - \log \left(\mathbb{E}_{x \sim \mathcal{D}} \mathbb{E}_{y \sim \pi_{\text{chosen}}(\cdot|x)} [e^{T_\phi(x, y)}] \right) \quad (\text{C.1})$$

$$\geq \sup_{\phi} \mathbb{E}_{x \sim \mathcal{D}, y^+ \sim \pi_{\text{chosen}}(\cdot|x)} [T_\phi(x, y^+)] - \log \left(\mathbb{E}_{x \sim \mathcal{D}} \{ \mathbb{E}_{y \sim \pi_{\text{chosen}}(\cdot|x)} [e^{T_\phi(x, y)}] + \mathbb{E}_{y \sim \pi_{\text{rejection}}(\cdot|x)} [e^{T_\phi(x, y)}] \} \right) \quad (\text{C.2})$$

$$= \sup_{\phi} \mathbb{E}_{x \sim \mathcal{D}, y^+ \sim \pi_{\text{chosen}}(\cdot|x)} [T_\phi(x, y^+)] - \log \left(\mathbb{E}_{x \sim \mathcal{D}} [2 \mathbb{E}_{y \sim \bar{\pi}(\cdot|x)} [e^{T_\phi(x, y)}]] \right) \quad (\text{C.3})$$

$$= \sup_{\phi} \mathbb{E}_{x \sim \mathcal{D}, y^+ \sim \pi_{\text{chosen}}(\cdot|x)} [T_\phi(x, y^+)] - \log \mathbb{E}_{x \sim \mathcal{D}, y \sim \bar{\pi}(\cdot|x)} [e^{T_\phi(x, y)}] - \log 2 \quad (\text{C.4})$$

$$= \sup_{\phi} \mathbb{E}_{x \sim \mathcal{D}, y^+ \sim \pi_{\text{chosen}}(\cdot|x)} [\log e^{T_\phi(x, y^+)}] - \log \mathbb{E}_{x \sim \mathcal{D}, y \sim \bar{\pi}(\cdot|x)} [e^{T_\phi(x, y)}] - \log 2. \quad (\text{C.5})$$

C.2. Let's Constrain T to a Restricted Critic Family

In the main text we further restrict the critic to a log-ratio family based on the fixed reference model π_{ref} (so that it can be evaluated with one additional forward pass through π_{ref}):

$$\mathcal{F}_{\text{sub}} = \left\{ T_\theta(x, y) = \log \frac{\pi_\theta(y | x)}{\pi_{\text{ref}}(y | x)} + c(x) \mid c : \mathcal{X} \rightarrow \mathbb{R} \right\}. \quad (\text{C.6})$$

(The additive offset $c(x)$ is irrelevant for pairwise differences and is included for generality.)

Plugging T_θ^{LR} into the modified DV objective yields a *surrogate* objective that depends only on θ :

$$\hat{I}(\theta) := \mathbb{E}_{x \sim \mathcal{D}, y^+ \sim \pi_{\text{chosen}}(\cdot|x)} [T_\theta(x, y^+)] - \log \mathbb{E}_{x \sim \mathcal{D}, y \sim \bar{\pi}(\cdot|x)} [e^{T_\theta(x, y)}] - \log 2. \quad (\text{C.7})$$

Since $\mathcal{F}_{\text{sub}} \subset \mathcal{F}_{\text{all}}$ and the mixture pool relaxation is a lower bound, we have

$$\hat{I}(\theta) \leq I_\pi^* \leq I(\pi_\theta, \pi_{\text{chosen}}). \quad (\text{C.8})$$

D. The JSD estimator yields substantially lower variance than DV/MINE

Subsequent research has shown that the DV-based objective used in MINE suffers from significant instability in practice (Poole et al., 2019). Specifically, its variance can grow exponentially with the true mutual information (MI) (Song & Ermon, 2020), leading to exploding gradients and requiring very large batch sizes to maintain stable training (Guo et al., 2022). Poole et al. (2019) further demonstrate that DV-style bounds like MINE exhibit low bias but extremely high variance, causing erratic gradient updates.

In the work (McAllester & Stratos, 2020), it is emphasized that accurate MI estimation is intrinsically challenging: high bias or high variance can lead to overfitting of spurious correlations. Thus, in contrastive learning, stable MI lower bounds (e.g., MINE or the Jensen–Shannon (JS) bound) should be prioritized over maximizing numerical MI estimates.

Additionally, Sordani et al. (2021) point out that the MINE bound is upper-bounded by $\log K$, where K is the number of negative samples. When the true MI greatly exceeds $\log K$, the bound severely underestimates MI.

Collectively, these studies (Guo et al., 2022; McAllester & Stratos, 2020) highlight that in regimes with few negative samples, MINE suffers from prohibitively large variance (Poole et al., 2019), leading to unstable gradients. Therefore, replacing MINE with a variance-reduced alternative, such as the JSD estimator (Wen et al., 2020), is crucial for reliable optimization.

We investigate the behavior of mutual information (MI) estimators in the low-negative-sample regime using a synthetic bivariate Gaussian setting, where the correlation coefficient ρ controls the dependency between variables. The ground-truth MI is given analytically as $I(X, Y) = -\frac{1}{2} \log(1 - \rho^2)$.

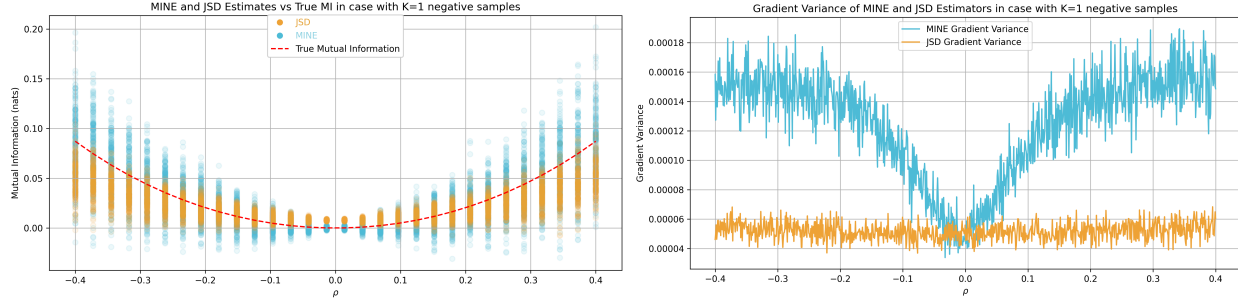


Figure 4. Left: Estimated mutual information under the bivariate Gaussian setting with varying ρ , using MINE and JSD estimators with $K = 1$ negative sample. While both methods track the ground-truth trend, JSD is consistently better than MINE. Right: Gradient variance of each estimator. The JSD estimator yields substantially lower variance, indicating more stable optimization behavior under limited negative sampling.

The left panel of Figure 4 compares the DV/MINE and JSD estimators with a single negative sample ($K = 1$). While both estimators qualitatively capture the ground-truth MI trend, JSD does not consistently outperform MINE in terms of accuracy. However, **when gradient stability is prioritized over precise MI estimation, JSD demonstrates a clear advantage** (Shrivastava et al., 2023). To further investigate this, we analyze the gradient variance induced by each estimator, with the right panel of Figure 4 showing the gradient variance computed from the first-layer weights of a three-layer MLP trained for each estimator’s objective. Under conditions with limited negative samples, **JSD exhibits significantly greater stability than MINE**, making it particularly suitable for training objectives, such as Alignment, that rely on contrastive MI estimates with sparse negative sampling (Chen et al., 2024).

E. Upper Bounds on the Jensen Gap

We consider the Jensen approximation used in deriving a KL-style objective:

$$\begin{aligned} & \mathbb{E}_{x \sim \mathcal{D}, y^+ \sim \pi_{\text{chosen}}(\cdot | x)} [T_\phi(x, y^+)] - \log \mathbb{E}_{x \sim \mathcal{D}, y \sim \pi_\theta(\cdot | x)} \left[\frac{\pi_\theta(y | x)}{\pi_{\text{chosen}}(y | x)} \right] \\ & \approx \mathbb{E}_{x \sim \mathcal{D}, y^+ \sim \pi_{\text{chosen}}(\cdot | x)} [T_\phi(x, y^+)] - \mathbb{E}_{x \sim \mathcal{D}, y \sim \pi_\theta(\cdot | x)} \left[\log \frac{\pi_\theta(y | x)}{\pi_{\text{chosen}}(y | x)} \right]. \end{aligned} \quad (\text{C.1})$$

For a fixed prompt x , define the positive random variable

$$f_x(Y) := \frac{\pi_\theta(Y | x)}{\pi_{\text{chosen}}(Y | x)}, \quad Y \sim \pi_\theta(\cdot | x),$$

and the corresponding Jensen gap

$$\Delta(x) := \mathbb{E}_{Y \sim \pi_\theta(\cdot | x)} [\log f_x(Y)] - \log \mathbb{E}_{Y \sim \pi_\theta(\cdot | x)} [f_x(Y)].$$

The variance-based bound. Let $\mu_x = \mathbb{E}_{Y \sim \pi_\theta(\cdot | x)} [f_x(Y)]$ and $\sigma_x^2 = \text{Var}_{Y \sim \pi_\theta(\cdot | x)} (f_x(Y))$. If $\sigma_x / \mu_x \ll 1$, then a second-order Taylor expansion of $\log$ around μ_x yields

$$\Delta(x) = \log \mu_x - \mathbb{E}[\log f_x] \leq \frac{\sigma_x^2}{2\mu_x^2}.$$

As the fluctuations of f_x decrease (i.e. $\sigma_x^2 \rightarrow 0$), the Jensen gap $\Delta(x)$ approaches zero. Because π_{chosen} is an energy-based re-weighting of π_{ref} (Eq. (A.1)),

$$\pi_{\text{chosen}}(y | x) = \pi_{\text{ref}}(y | x) \frac{\exp(\alpha r(x, y))}{Z(x)},$$

we can rewrite the ratio as

$$f_x(y) = \frac{\pi_{\theta}(y | x)}{\pi_{\text{chosen}}(y | x)} = Z(x) e^{-\alpha r(x, y)} \cdot \frac{\pi_{\theta}(y | x)}{\pi_{\text{ref}}(y | x)}.$$

Equivalently,

$$\log f_x(y) = \log \frac{\pi_{\theta}(y | x)}{\pi_{\text{ref}}(y | x)} + \log Z(x) - \alpha r(x, y).$$

Therefore, keeping the likelihood ratio $\frac{\pi_{\theta}(\cdot | x)}{\pi_{\text{ref}}(\cdot | x)}$ close to unity reduces the variability of f_x (hence σ_x^2) and tightens the Jensen approximation, providing an alternative justification for ratio-clipping heuristics in PPO-style methods.

F. Preliminaries

Notation Let $x \in \mathcal{X}$ be a user prompt, $y \in \mathcal{Y}$ a textual response, and $\pi(y | x)$ a conditional language model (policy). We denote

- π_{ref} — a fixed reference model (e.g. SFT checkpoint),
- π_{θ} — the trainable policy with parameters θ ,
- π_{chosen} — the human-preferred generator producing positive samples y^+ , modeled as an energy-based model (see Equation A.1).
- $\pi_{\text{rejection}}$ — the lower-preference generator producing negative samples y^- , also instantiated as an energy-based model.
- $\bar{\pi} = \frac{1}{2}(\pi_{\text{chosen}} + \pi_{\text{rejection}})$ — the mixed negative pool used in MINE sampling.

Reinforcement Learning from Human Feedback. Given the estimated reward function $r(\mathbf{x}, \mathbf{y})$, dictating the human preferences, RLHF fine-tunes policy π_{θ} by optimizing the following objective:

$$\max_{\pi_{\theta}} \mathbb{E}_{\mathbf{y} \sim \pi_{\theta}(\mathbf{y} | \mathbf{x})} [r(\mathbf{x}, \mathbf{y})] - \beta \text{D}_{\text{KL}} [\pi_{\theta}(\mathbf{y} | \mathbf{x}) \| \pi_{\text{ref}}(\mathbf{y} | \mathbf{x})], \quad (\text{F.1})$$

where $\beta > 0$ is an appropriate KL penalty coefficient. RLHF typically optimizes the above objective in Equation F.1 using RL algorithms, such as PPO (Schulman et al., 2017). Although RLHF with PPO has achieved remarkable success, the training process of PPO is unstable because of the high variance of the estimates of the policy gradients (Engstrom et al., 2020).

Reward Modeling. One standard approach to reward modeling is to fit a reward function $r_{\phi}(\mathbf{x}, \mathbf{y})$ with the BT preference model in Equation (1). Specifically, the reward function $r_{\phi}(\mathbf{x}, \mathbf{y})$ can be estimated by maximizing the log-likelihood over preference feedback $(\mathbf{x}, \mathbf{y}^+, \mathbf{y}^-)$:

$$\mathcal{L}_{\text{RM}}(\phi; \mathcal{D}) = \mathbb{E}_{(\mathbf{x}, \mathbf{y}^+, \mathbf{y}^-) \sim \mathcal{D}} [-\log \sigma(r_{\phi}(\mathbf{x}, \mathbf{y}^+) - r_{\phi}(\mathbf{x}, \mathbf{y}^-))]. \quad (\text{F.2})$$

Mutual Information and the DV Lower Bound For two random variables (Y, Z) with joint P_{YZ} and marginals P_Y, P_Z , the mutual information is $I(Y; Z) = \text{KL}(P_{YZ} \| P_Y P_Z)$. The **Donsker–Varadhan** (DV) variational form writes

$$I(Y; Z) = \sup_{T \in \mathcal{F}} \left(\mathbb{E}_{P_{YZ}}[T] - \log \mathbb{E}_{P_Y P_Z}[e^T] \right), \quad (\text{F.3})$$

where $\mathcal{F}$ is any function class with finite log-moment (Donsker & Varadhan, 1983; Belghazi et al., 2018b). Belghazi et al. (2018a) propose parametrising T_{ϕ} by a neural network and maximising the RHS of equation F.3; the resulting estimator is known as *MINE*.

Energy-based Models. Energy-based models (EBMs) (Lecun et al., 2006) define the distribution through an energy function. For $\mathbf{y} \in \mathbb{R}^D$, its probability density can be expressed as follows:

$$p_{\theta}(\mathbf{y}) = \exp(-E_{\theta}(\mathbf{y}))/Z_{\theta}(\mathbf{y}), \quad (\text{F.4})$$

where $E_{\theta}(\mathbf{y}) : \mathbb{R}^D \rightarrow \mathbb{R}$ is the energy function, mapping the data point $\mathbf{y}$ to a scalar, and $Z_{\theta}(\mathbf{y}) = \sum_{\mathbf{y}} \exp(-E_{\theta}(\mathbf{y}))$ is the unknown normalization constant (Song & Kingma, 2021).

G. Experimental Details

The source code used in our experiments is available at: <https://anonymous.4open.science/r/MIO-63E6/>

G.1. Training Details

Table 8. The hyperparameter search space for the baselines.

Method	Method
DPO $\beta \in [0.01, 0.05, 0.1]$	IPO $\tau \in [0.01, 0.1, 0.5, 1.0]$
NCA $\beta \in [0.01, 0.10, 0.15]$	SLiC $\lambda \in [0.1, 0.5, 1.0, 10.0], \delta \in [0.1, 0.5, 1.0, 2.0]$
KTO $\lambda_l = \lambda_w = 1.0$	SimPO $\beta \in [2.0, 2.5]$
$\beta \in [0.01, 0.05, 0.1]$	$\gamma \in [0.3, 0.5, 1.0, 1.2, 1.4, 1.6]$
DIL $\beta \in [0.8, 1.0, 1.2]$	ORPO $\beta \in [0.01, 0.05, 0.1]$
CPO $\beta \in [0.1, 0.01]$	MIO $\beta \in [5.0, 7.5, 8.75, 10.0]$

We follow the general training configurations established in SimPO. During the supervised fine-tuning (SFT) stage, we use a learning rate of 2×10^{-5} . For both SFT and preference optimization, we set the batch size to 128 and the maximum sequence length to 2048. A cosine learning rate scheduler is applied with a 10% warm-up ratio over a single epoch. We use the Adam optimizer (Kingma & Ba, 2014) in all experiments.

Method-specific hyperparameters are selected according to the search strategy described in SimPO. Each baseline method uses its own dedicated hyperparameter set, as detailed in Table 8. The learning rate for each method is tuned within the range $\{3\text{e-}7, 5\text{e-}7, 7\text{e-}7, 1\text{e-}6, 5\text{e-}6\}$.

To mitigate length bias, we normalize the response likelihood by computing the average log-probability of all tokens in the response under the policy model, following SimPO and DIL.

The mistral-7b-sft experiments are conducted using six NVIDIA A6000 GPUs (46GB each), with a total batch size of 384, while the llama and qwen experiments were performed on eight NVIDIA H20 GPUs. We build on the publicly available `alignment-handbook` codebase.

G.2. The Details of Datasets

UltraFeedbackBinarized (Cui et al., 2024; Tunstall et al., 2023) is a dataset comprising 64k prompts, each paired with four completions generated by various open-source and proprietary language models. GPT-4 assigns scores to these completions based on criteria such as helpfulness, honesty, and other qualitative metrics. To form binary preference pairs, the completion with the highest average score is selected as the preferred response, while one of the remaining three completions is randomly chosen as the rejected response.

G.3. Evaluation Tasks: Math and Reasoning Benchmarks

This section introduces the benchmark used for model evaluation. We follow prior work to evaluate the model fine-tuned on the UltraFeedbackBinarized dataset. Subsequently, we assess its performance on a suite of eight math- and reasoning-oriented tasks, aiming to comprehensively evaluate its quantitative reasoning ability, multi-step problem-solving skills, and domain-specific competence.

HendrycksMath (Hendrycks et al., 2021b) consists of 12,500 high school mathematics problems spanning algebra, geometry, and number theory. Each problem is accompanied by a step-by-step solution, targeting rigorous symbolic reasoning.

MinervaMath (Lewkowycz et al., 2022b) evaluates quantitative reasoning over STEM domains using university-level mathematics, science, and engineering problems. It emphasizes complex problem solving in structured formats.

MultiMedQA (Singhal et al., 2023) is a multi-task benchmark for medical question answering, aggregating datasets from professional exams, clinical research, and consumer health queries. It measures both factual correctness and clinical alignment.

MathQA (Amini et al., 2019b) contains over 37k multiple-choice math word problems paired with symbolic execution programs. The dataset encourages interpretable problem solving using operation-based formalisms.

GSM8K (Cobbe et al., 2021b) includes 8,500 elementary school math word problems requiring multi-step reasoning. It is widely used to benchmark arithmetic and logical problem-solving capabilities of language models. For this task, we report results using the `exact match` metric under the `flexible-extract` setting, which allows for variations in answer formatting.

AQuA-RAT (Cui et al., 2023) focuses on algebraic word problems from the AQuA-RAT dataset, requiring rationales and supporting steps. It is part of the broader AGIEval benchmark designed to test cognitive ability in foundation models. For this task, we use `normalized accuracy (acc_norm)` as the evaluation metric to account for formatting variability in symbolic responses.

MATH-Hard is a curated subset from the Hugging Face Open LLM Leaderboard that emphasizes challenging math problems, including multi-hop symbolic reasoning and advanced problem types.

MuSR (Sprague et al., 2023) targets multistep soft reasoning problems, typically presented in narrative forms such as logic puzzles and commonsense tasks. It evaluates language models’ ability to track state, causality, and inference chains over extended contexts.

H. Proof of DV/MINE Starvation Theorem

H.1 Setup and assumptions

Objective. We adopt the DV/MINE lower bound and the RLHF sampling scheme used in the paper. *DV/MINE objective (up to an additive constant):*

$$I_{DV}(\theta) = \sup_{\phi} \mathbb{E}_{x \sim \mathcal{D}, y^+ \sim \pi_{\text{chosen}}(\cdot|x)} [T_{\phi}(x, y^+)] - \log \left(\mathbb{E}_{x \sim \mathcal{D}} \mathbb{E}_{y \sim \bar{\pi}(\cdot|x)} e^{T_{\phi}(x, y)} \right) \quad (\text{H.1})$$

This is the same modified DV bound and sampling as in Appendix C.1 of the paper.

We will consider two practically important critic families T_{ϕ} :

θ -independent T : Here, T_{ϕ} is independent of θ . We train a separate neural network ϕ to estimate the value, which is equivalent to training a reward model in RLHF.

Log-ratio restricted T function:

$$T_{\theta}(x, y) = \log \frac{\pi_{\theta}(y|x)}{\pi_{\text{ref}}(y|x)} + c(x), \quad (\text{H.2})$$

for some (irrelevant) constant offset c . This coincides with the restricted family $\mathcal{F}_{\text{sub}}$ in Appendix C.4 and with the derivations leading to DPO in §2/B.1 of the main text.

Support assumption. We also use the paper’s support assumption (Appendix A): π_{chosen} and $\pi_{\text{rejection}}$ are energy-based re-weightings of π_{ref} (Eq. (A.1)), hence

$$\pi_{\text{ref}}(y^* | x^*) = 0 \implies \pi_{\text{chosen}}(y^* | x^*) = \pi_{\text{rejection}}(y^* | x^*) = \bar{\pi}(y^* | x^*) = 0. \quad (\text{H.3})$$

Softmax parameterization and the “own-logit” coordinate. Throughout, we consider the softmax/logit parameterization: for every prompt x , there are logits $s_{\theta}(x, y)$ such that

$$\pi_{\theta}(y | x) = \frac{e^{s_{\theta}(x, y)}}{\sum_{y'} e^{s_{\theta}(x, y')}}. \quad (\text{H.4})$$

Let u denote the particular logit coordinate $u := s_\theta(x^*, y^*)$. Differentiating the log-softmax gives

$$\frac{\partial}{\partial u} \log \pi_\theta(y | x) = \begin{cases} 1 - \pi_\theta(y^* | x^*), & \text{if } (x, y) = (x^*, y^*), \\ -\pi_\theta(y^* | x^*), & \text{if } x = x^*, y \neq y^*, \\ 0, & \text{if } x \neq x^*. \end{cases} \quad (\text{H.5})$$

We analyze the directional derivative along the score direction $\nabla_\theta \log \pi_\theta(y^* | x^*)$. Under mild regularity (local reparameterization), this is proportional to the partial derivative w.r.t. the “own-logit” $u = s_\theta(x^*, y^*)$. Hence it suffices to compute $\partial I_{\text{DV}}(\theta)/\partial u$; the claimed inner product equals this derivative up to a positive scaling.

H.2 Main theorem: DV/MINE Starvation Theorem

Theorem H.1 (Gradient starvation for DV/MINE). Fix (x^*, y^*) with $\pi_{\text{ref}}(y^* | x^*) = 0$ so that (H.3) holds. Consider $I_{\text{DV}}(\theta)$ in (H.1). Then for any θ with $\pi_\theta(y^* | x^*) \geq 0$ (in particular at the boundary $\pi_\theta(y^* | x^*) = 0$),

$$\langle \nabla_\theta I_{\text{DV}}(\theta), \nabla_\theta \log \pi_\theta(y^* | x^*) \rangle = 0,$$

in each of the following two cases:

- (a) T_ϕ does not depend on θ (standard MINE critic), or
- (b) $T_\theta(x, y) = \log \frac{\pi_\theta(y|x)}{\pi_{\text{ref}}(y|x)} + c$ (log-ratio restricted critic from Appendix C.2).

Proof. We differentiate $I_{\text{DV}}(\theta)$ w.r.t. the logit $u = s_\theta(x^*, y^*)$ and treat the two cases separately. By the envelope theorem (the inner $\sup_\phi$ can be ignored when differentiating w.r.t. θ at the maximizer), the derivative equals the θ -partial of the objective at the active critic. This yields

$$\frac{\partial I_{\text{DV}}(\theta)}{\partial u} = \underbrace{\mathbb{E}_{P_{\pi_\theta \pi_{\text{chosen}}}} \left[\frac{\partial T}{\partial u} \right]}_{(A)} - \underbrace{\frac{\mathbb{E}_{P_{\pi_\theta P_\pi}} [e^T \cdot \frac{\partial T}{\partial u}]}{\mathbb{E}_{P_{\pi_\theta P_\pi}} [e^T]}}_{(B)}, \quad (\text{H.6})$$

Case (a): θ -independent critic. Then $\partial T/\partial u \equiv 0$, hence both (A) and (B) vanish, and $\partial I_{\text{DV}}/\partial u = 0$. Therefore the claimed inner product is 0.

Case (b): T is a log-ratio critic. Here

$$\frac{\partial T}{\partial u} = \frac{\partial T}{\partial u}(\pi_\theta, \pi_{\text{chosen}}) = \frac{\partial}{\partial u} \log \pi_\theta(y | x) = \mathbf{1}\{x = x^*\}(\mathbf{1}\{y = y^*\} - \pi_\theta(y^* | x^*)) \quad \text{by (H.5)}. \quad (\text{H.7})$$

Using (H.7) in (A) and the fact $\pi_{\text{chosen}}(y^* | x^*) = 0$ (H.3), we get

$$(A) = \mathbb{E}_{x \sim D} \mathbf{1}\{x = x^*\} \mathbb{E}_{y^+ \sim \pi_{\text{chosen}}(\cdot | x^*)} [\mathbf{1}\{y^+ = y^*\} - \pi_\theta(y^* | x^*)] = -\pi_\theta(y^* | x^*). \quad (\text{H.8})$$

Similarly, using (H.7) in (B) and $\bar{\pi}(y^* | x^*) = 0$ (H.3),

$$(B) = \frac{\mathbb{E}_{x \sim D} \mathbf{1}\{x = x^*\} \mathbb{E}_{y \sim \bar{\pi}(\cdot | x^*)} [e^{T(x^*, y)} (\mathbf{1}\{y = y^*\} - \pi_\theta(y^* | x^*))]}{\mathbb{E}_{x \sim D} \mathbf{1}\{x = x^*\} \mathbb{E}_{y \sim \bar{\pi}(\cdot | x^*)} [e^{T(x^*, y)}]} = -\pi_\theta(y^* | x^*). \quad (\text{H.9})$$

Plugging (H.8)–(H.9) into (H.6) gives $\frac{\partial I_{\text{DV}}}{\partial u} = (-\pi_\theta) - (-\pi_\theta) = 0$. Again, the claimed inner product is zero. $\square$

H.3 Quantitative decay near the boundary

Theorem H.1 shows exact cancellation for the two common critic forms. One may also ask for quantitative bounds when the critic depends on θ more generally.

Theorem H.2 (Linear decay near zero under Lipschitz critics). Suppose the active critic can be written as $T_\theta(\pi_\theta, \pi_{\text{chosen}}) = G(x, y, \log \pi_\theta(y | x))$ where G is Lipschitz in its third argument with constant L (uniformly in (x, y)). Under the support condition (H.3), for any θ ,

$$|\langle \nabla_\theta I_{\text{DV}}(\theta), \nabla_\theta \log \pi_\theta(y^* | x^*) \rangle| \leq 2L \pi_\theta(y^* | x^*). \quad (\text{H.10})$$

In particular, as $\pi_\theta(y^* | x^*) \downarrow 0$, the directional derivative vanishes at least linearly.

Proof. As in (H.6), $\partial I_{DV} / \partial u = \mathbb{E}_{P_{\pi_\theta} \pi_{\text{chosen}}} [\partial_u T_\theta] - \frac{\mathbb{E}_{P_{\pi_\theta} P_\pi} [e^{T_\theta} \partial_u T_\theta]}{\mathbb{E}_{P_{\pi_\theta} P_\pi} [e^{T_\theta}]}$. By the chain rule, $\partial_u T_\theta = (\partial_3 G) \cdot \partial_u \log \pi_\theta(y | x)$. The Lipschitz assumption yields $|\partial_3 G| \leq L$ almost everywhere; combine this with (H.5) and (H.3) to get

$$|\mathbb{E}_{P_{\pi_\theta} \pi_{\text{chosen}}} [\partial_u T_\theta]| \leq L \mathbb{E}_x \mathbf{1}\{x = x^*\} \mathbb{E}_{y^+ \sim \pi_{\text{chosen}}(\cdot | x^*)} [|\mathbf{1}\{y^+ = y^*\} - \pi_\theta(y^* | x^*)|] = L \pi_\theta(y^* | x^*),$$

and similarly

$$\left| \frac{\mathbb{E}_{P_{\pi_\theta} P_\pi} [e^{T_\theta} \partial_u T_\theta]}{\mathbb{E}_{P_{\pi_\theta} P_\pi} [e^{T_\theta}]} \right| \leq L \pi_\theta(y^* | x^*).$$

Triangle inequality then implies (H.10). $\square$

Consequences. For the log-ratio critic (Theorem H.1(b)), we have $L = 1$ and the two terms are exactly equal, yielding exact zero (stronger than (H.10)). For any θ -independent critic (Theorem H.1(a)), $\partial_3 G \equiv 0$, hence the bound is trivially zero. For other smooth θ -dependent critics, (H.10) shows a linear decay of the directional derivative as $\pi_\theta(y^* | x^*) \rightarrow 0$, formalizing the “the closer to the boundary, the weaker the signal” intuition.

I. MIO leverages the reward signal more effectively than NCA-Pair.

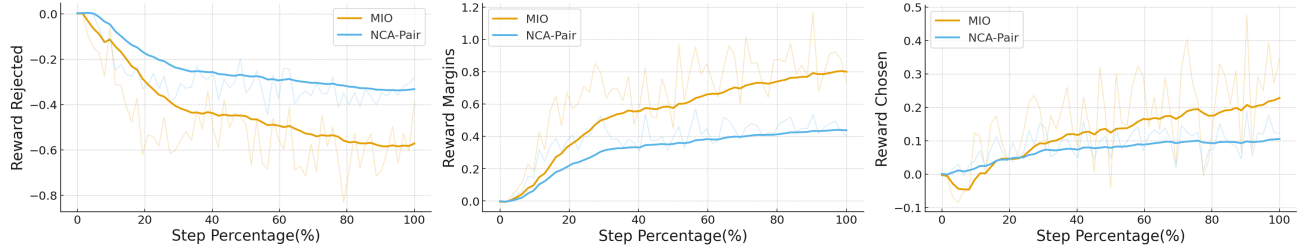


Figure 5. MIO leverages the reward signal more effectively than NCA-Pair.

Figure 5 presents a direct comparison between MIO and NCA-Pair in terms of reward margin and related metrics(sentence level). The results show that MIO achieves a greater increase in both the chosen reward and the reward margin, as well as a larger decrease in the rejected reward, compared to NCA-Pair.

J. LLM Usage

Large Language Models (LLMs) were used solely to assist with writing and editing of this manuscript—for tasks such as wording refinement, grammar checking, and improving readability. LLMs were not involved in research ideation, methodology, data collection, analysis, or result selection. All scientific content and claims were produced and verified by the authors, who take full responsibility for the manuscript. We ensured that any LLM-assisted text complies with ethical guidelines and does not constitute plagiarism or scientific misconduct.